

CogAge multi-subject optimizer comparison

Dataset, fixed model, and split policy

Four labels are held fixed across subjects: Bending=0, Walking=1, Sitting=2, Standing=3. Only subjects with all four activities in both original splits are included in four-class experiments. The inspected prepared datasets include all S1-S8: S1, S2, S5, S7 are complete; S3, S6, S8 are partial; S4 has no original test folders. Partial subjects are previewed but not assigned fabricated classes.

Original_1_extracted folders supply training windows; _2_extracted folders supply held-out testing windows. A seeded 20% Event-group validation split is made within each class from the original training folders only. No Event crosses model-train and validation. The model scaler is refitted on model-train windows and then applied to validation and test. Test metrics use the best validation-loss checkpoint and are not used for tuning.

Input: 128 x 33 flattened (4224); network 4224 -> 32 -> 16 -> 4 logits; ReLU hidden layers; 135,796 parameters; stable sparse cross-entropy from logits. NumPy implementation; He-normal initialization seed 42; 100 epochs; learning rate 0.01; mini-batch 8; momentum 0.9; AdamW decoupled weight decay 0.01. Full-batch, mini-batch and stochastic GD update once per full set, mini batch and example respectively; the five adaptive/momentum configurations use mini batches.

Per-subject class and split counts

S1: model train 32 [Bending:8, Walking:8, Sitting:8, Standing:8]; validation 8 [Bending:2, Walking:2, Sitting:2, Standing:2]; test 40 [Bending:10, Walking:10, Sitting:10, Standing:10] | S2: model train 32 [Bending:8, Walking:8, Sitting:8, Standing:8]; validation 8 [Bending:2, Walking:2, Sitting:2, Standing:2]; test 40 [Bending:10, Walking:10, Sitting:10, Standing:10] | S5: model train 32 [Bending:8, Walking:8, Sitting:8, Standing:8]; validation 8 [Bending:2, Walking:2, Sitting:2, Standing:2]; test 40 [Bending:10, Walking:10, Sitting:10, Standing:10] | S7: model train 30 [Bending:8, Walking:8, Sitting:6, Standing:8]; validation 8 [Bending:2, Walking:2, Sitting:2, Standing:2]; test 30 [Bending:6, Walking:6, Sitting:9, Standing:10]

Subject	Optimizer	Best val loss	Epoch	Test acc	Macro F1	Seconds	Updates
S1	Mini-batch GD	1.125	2	0.800	0.783	9.35	400
S1	Stochastic GD	0.002	1	0.825	0.827	7.87	3,200
S1	SGD + Momentum	1.545	1	0.825	0.791	8.70	400
S1	Adagrad	0.615	2	0.900	0.899	6.46	400
S1	RMSprop	4.477	1	0.800	0.799	6.46	400
S1	Adam	0.764	4	0.925	0.925	7.59	400
S1	AdamW	0.756	4	0.925	0.925	8.16	400
S2	Full-batch GD	2.106	3	0.750	0.683	5.15	100
S2	Mini-batch GD	1.859	4	0.800	0.757	11.84	400
S2	Stochastic GD	0.502	3	0.975	0.975	7.59	3,200
S2	SGD + Momentum	1.282	3	0.825	0.825	22.87	400
S2	Adagrad	0.043	1	0.950	0.948	9.08	400
S2	RMSprop	3.454	5	0.900	0.896	14.78	400
S2	Adam	2.242	1	0.900	0.893	7.86	400
S2	AdamW	2.240	1	0.900	0.893	9.00	400
S5	Full-batch GD	0.618	85	0.800	0.750	2.86	100
S5	Mini-batch GD	0.176	74	0.775	0.705	5.30	400
S5	Stochastic GD	0.571	6	0.875	0.875	5.85	3,200
S5	SGD + Momentum	0.000	10	0.750	0.661	4.74	400
S5	Adagrad	0.054	100	0.850	0.849	4.79	400
S5	RMSprop	0.000	2	0.625	0.577	7.68	400
S5	Adam	0.006	4	0.750	0.723	10.64	400
S5	AdamW	0.006	4	0.750	0.723	10.85	400
S7	Full-batch GD	0.138	100	0.615	0.540	4.10	100
S7	Mini-batch GD	0.057	100	0.667	0.588	15.02	400
S7	Stochastic GD	0.000	100	0.744	0.661	7.08	3,000
S7	SGD + Momentum	0.063	1	0.641	0.572	5.99	400
S7	Adagrad	0.000	100	0.718	0.642	7.41	400
S7	RMSprop	-0.000	1	0.744	0.696	11.30	400
S7	Adam	0.000	3	0.744	0.661	8.36	400
S7	AdamW	0.000	3	0.744	0.661	7.66	400

Validation loss by epoch

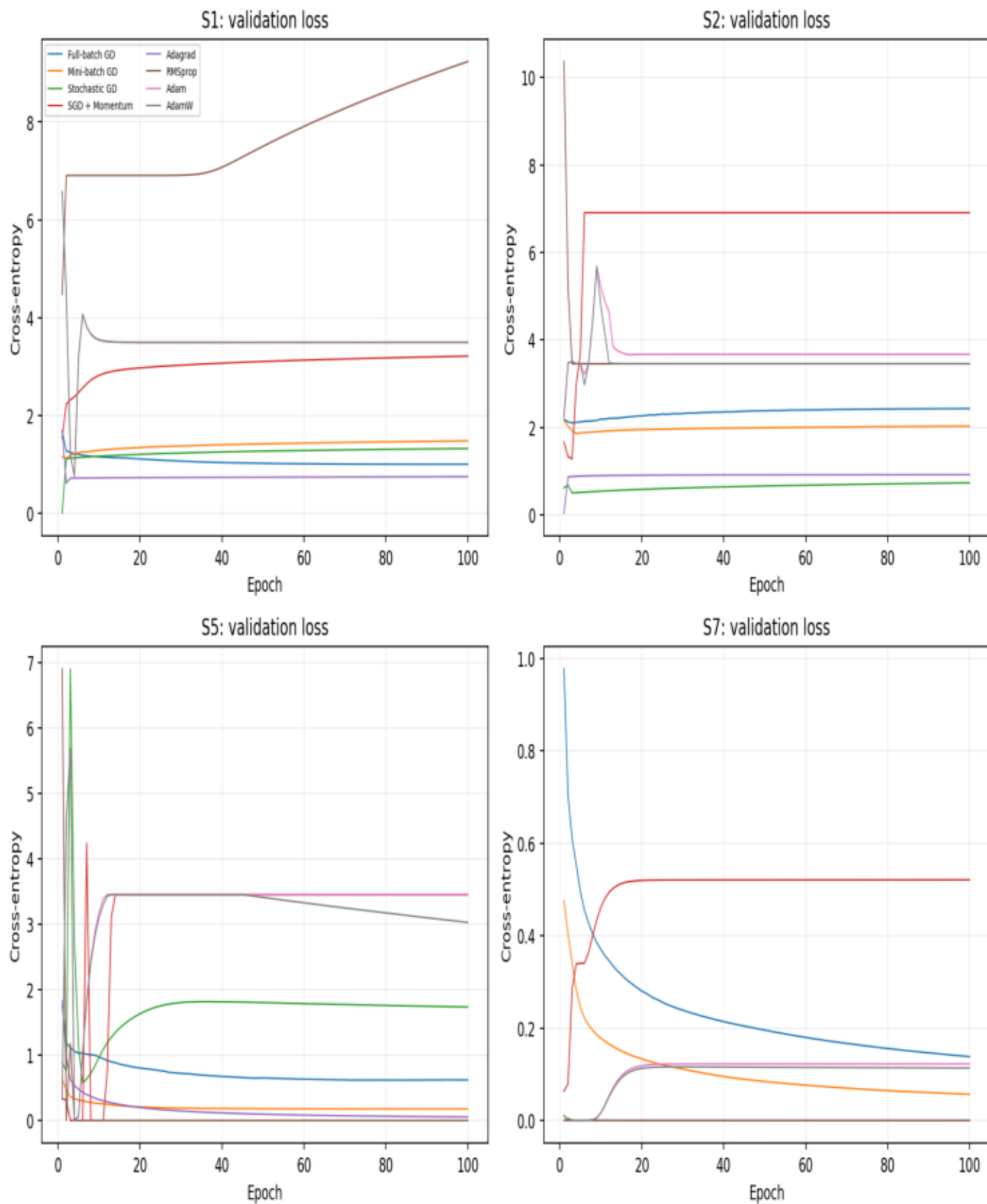


Figure 1. Validation loss for all eight optimizer runs, shown separately for each complete subject. Checkpoints are selected using validation loss.

Validation accuracy by epoch

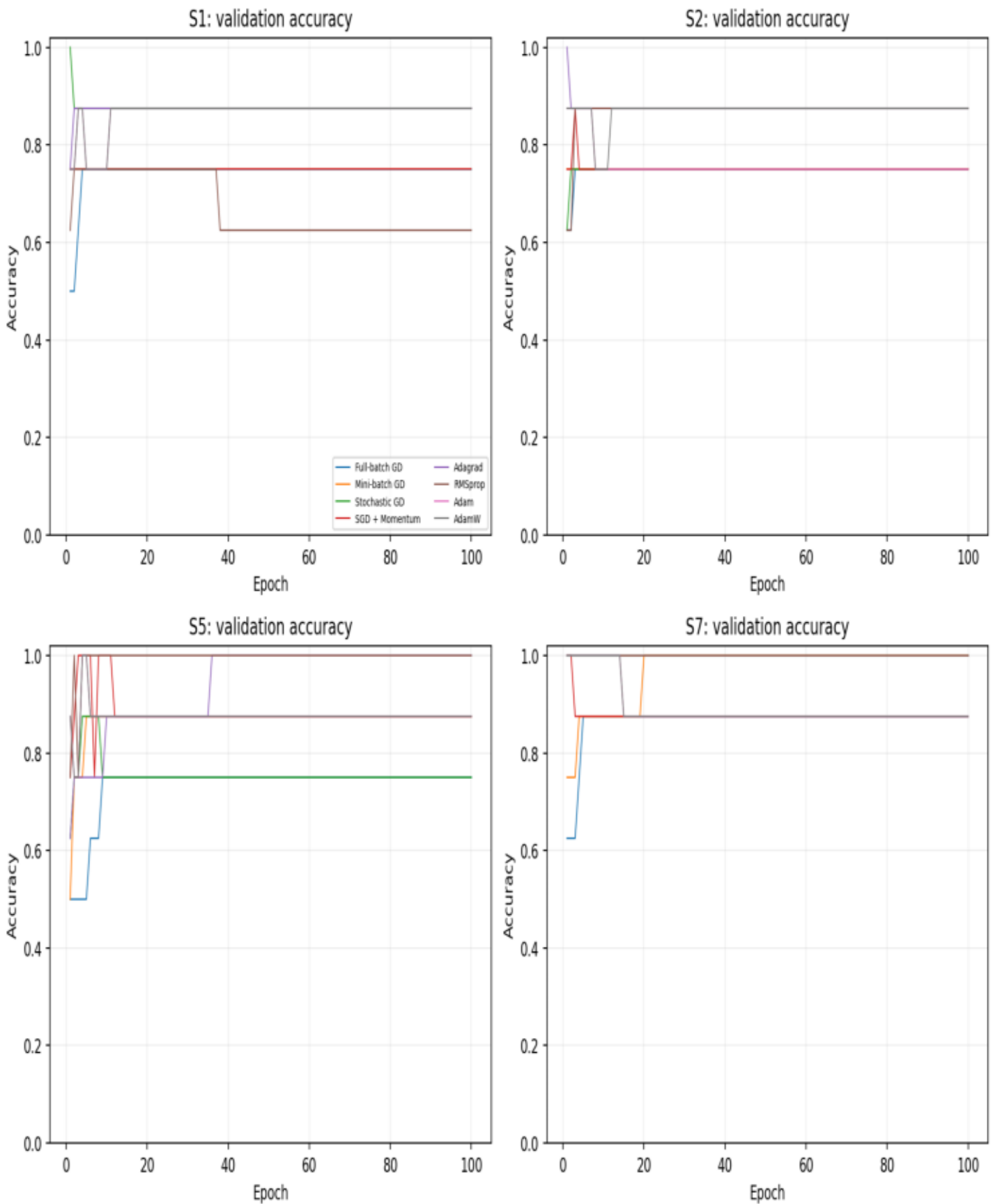


Figure 2. Validation accuracy for each optimizer and subject. Validation contains few Events, so a single misclassification changes accuracy substantially.

Convergence and test comparisons

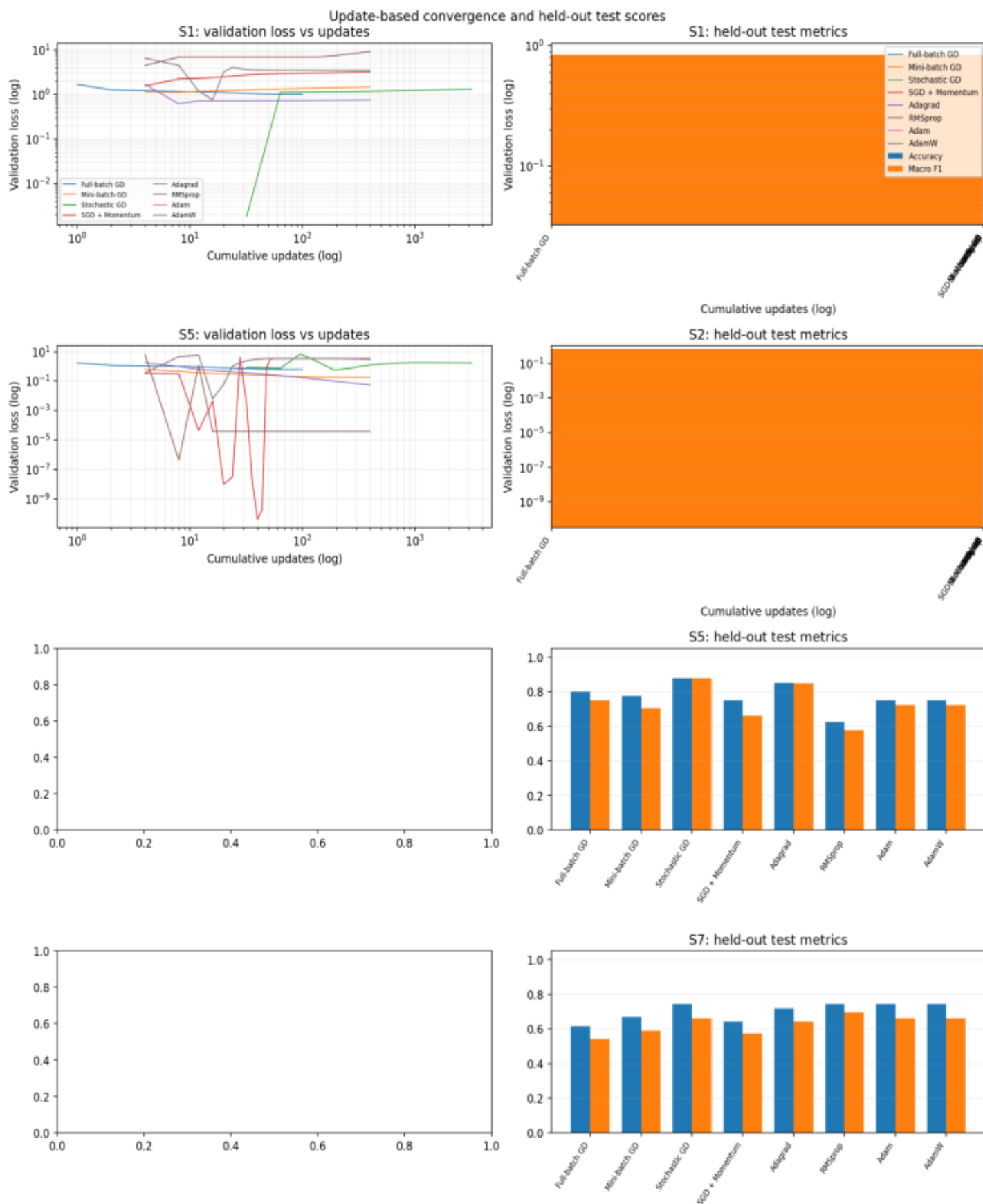


Figure 3. Left panels compare validation loss against cumulative updates; right panels show test accuracy and macro F1. Epochs alone are not comparable because batch sizes produce different update counts.

Confusion matrices: S1, S2

Held-out test confusion matrices: S1, S2

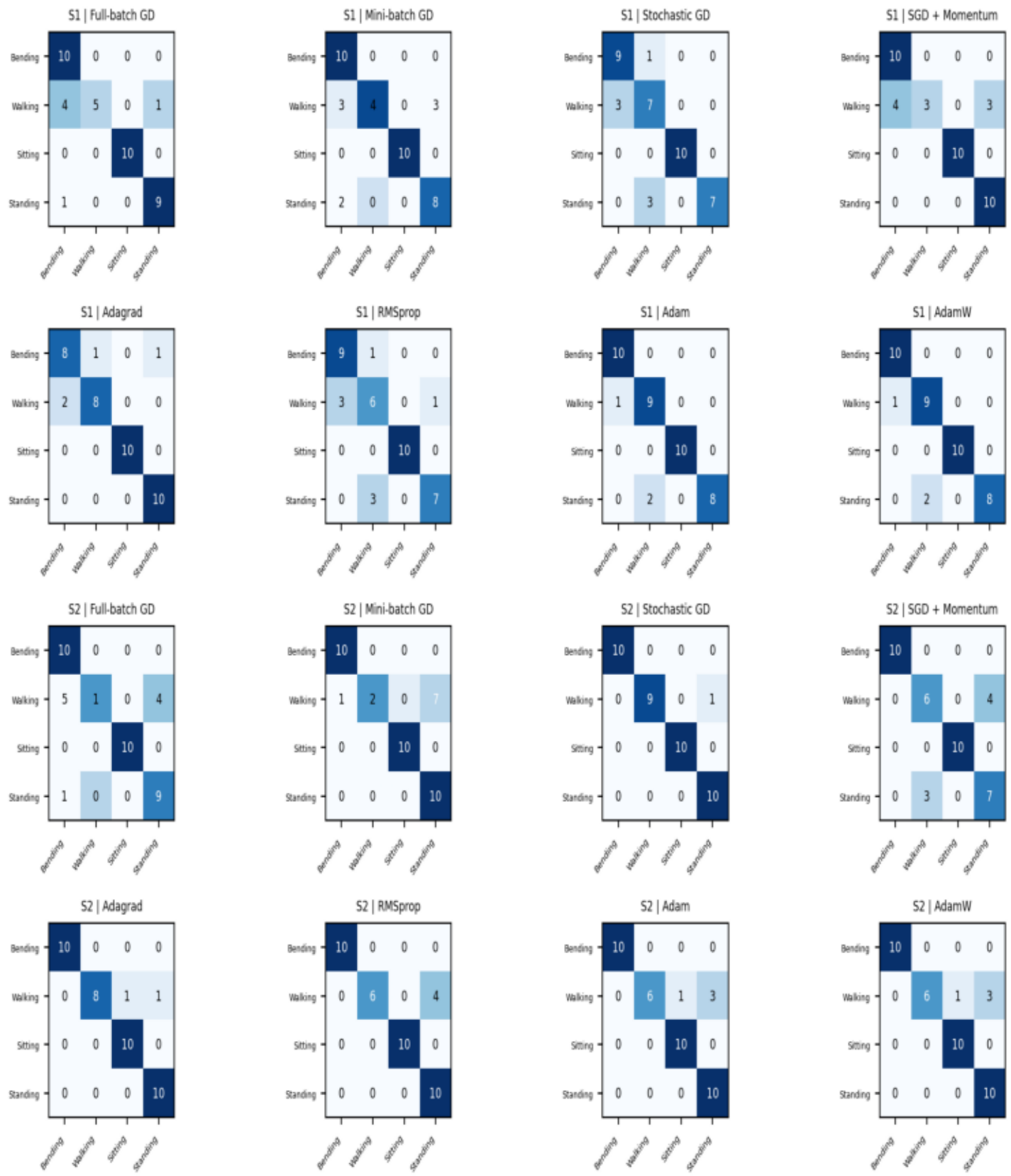


Figure 4. One 4 x 4 test confusion matrix per optimizer. Rows are true labels and columns are predicted labels; class order is Bending, Walking, Sitting, Standing.

Confusion matrices: S5, S7

Held-out test confusion matrices: S5, S7

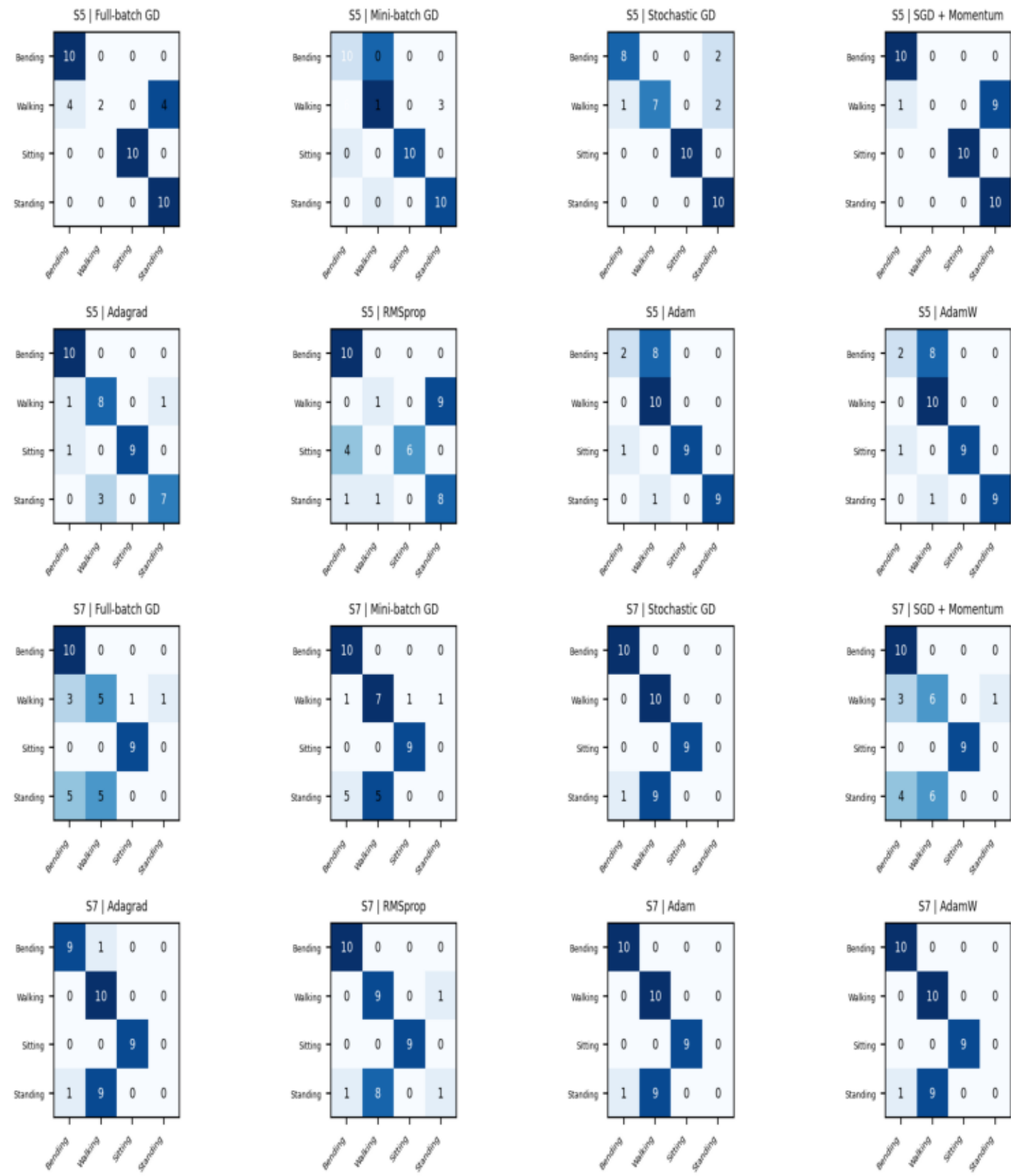


Figure 4. One 4 x 4 test confusion matrix per optimizer. Rows are true labels and columns are predicted labels; class order is Bending, Walking, Sitting, Standing.